PROJECT: AUTOMATING DATA IMPORT AND RELATIONSHIP MAPPING USING IMPORT SETS & DOT WALKING

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PROJECT REPORT

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1. INTRODUCTION

1.1 Project Overview

In modern organizations, large amounts of data are continuously generated from multiple sources such as spreadsheets, databases, and external applications. Managing and importing this data into a centralized system can be challenging, especially when maintaining proper relationships between different tables and records. Manual data import processes often require significant effort, time, and careful monitoring to ensure that the data is accurate and properly linked. This project focuses on **automating the data import process** using **Import Sets and Dot Walking**. Import Sets allow data to be temporarily stored and processed before it is inserted into the main tables. Transform Maps are then used to map the incoming data fields to the appropriate fields in the target tables. Dot Walking enables easy access and linking of related records across tables, helping to maintain proper relationships within the database. By automating these processes, the system can efficiently handle large volumes of data while reducing manual intervention. The use of Import Sets ensures that data is validated and transformed before it enters the system, while Dot Walking simplifies the process of accessing related information across multiple tables.

1.2 Purpose of the Project

The main purpose of this project is to **simplify and automate the data import process** while maintaining accurate relationships between different data tables. Many organizations face difficulties when handling large datasets because manual processes often lead to errors, duplication, and inconsistencies. This project aims to solve these challenges by implementing an automated solution that improves efficiency and data accuracy. The use of Import Sets helps manage incoming data effectively, while Dot Walking ensures that relationships between records are properly maintained without complex manual mapping. Additionally, this project helps improve system performance by reducing processing time and minimizing human errors. By automating data imports and relationships, organizations can maintain a reliable and well-structured database that supports better decision-making and system operations.

2. IDEATION PHASE

2.1Brainstorming & Idea Prioritization

Step 1 – Brainstormed Ideas

- Manual data import using scripts
- Import Sets with manual relationship mapping
- Automated Import Sets with Transform Maps
- Import automation using Dot Walking

Step 2 – Grouping

- Manual Solutions
- Script-based data import
- Semi-Automated Solutions
- Import Sets with manual field mapping
- Fully Automated Solutions
- Import Sets with Transform Maps
- Dot Walking for automatic relationship mapping

Step 3 – Prioritization

Selected Solution:

- Automating Data Import using Import Sets and Dot Walking

Reason:

- High feasibility
- High efficiency
- Reduces manual effort
- Maintains accurate relationships between tables

2.2 Define Problem Statement

PROBLEM STATEMENT- 1 : **Data Administrator**

I am: Data Administrator
I'm trying to: Import large volumes of data into the system efficiently
But: Data relationships between tables are difficult to maintain
Because: Data comes from multiple external sources and requires manual mapping
Which makes me feel: Frustrated and the process becomes time-consuming and error-prone

PROBLEM STATEMENT -2: **System Manager**

I am: System Manager
I'm trying to: Maintain accurate and reliable data in the system
But: Manual data import and relationship mapping often creates errors
Because: Data fields are not automatically linked across tables
Which makes me feel: Uncertain about data accuracy and system reliability

IDEATION PHASE

Problem Statements (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because
PS-1	Data Administrator	import large volumes of data into the system	data relationships are difficult	data is coming from multiple external sources
PS-2	System Manager	maintain accurate data relationships	manual mapping creates errors	data fields are not automatically linked

2.3 Empathy Map Canvas

User Persona: Data Administrator

Think & Feel

- Worried about incorrect data relationships
- Concerned about data inconsistency

Hear

- Complaints from system users
- Feedback from management about data errors

See

- Multiple spreadsheets and databases
- Large amounts of raw imported data

Say & Do

- Manually map fields
- Check data relationships repeatedly

Pain

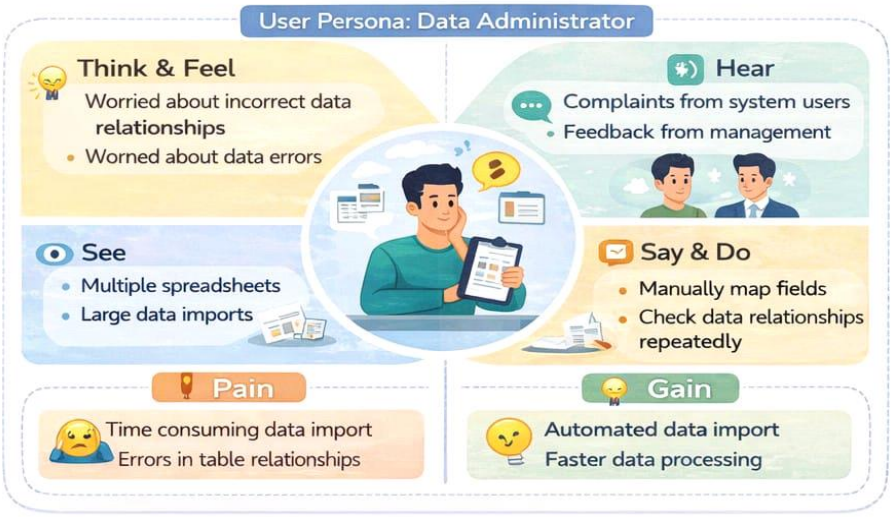
- Time-consuming data import
- Errors in relationships between tables

Gain

- Automated data import
- Accurate relationships using dot walking
- Faster data processing

IDEATION PHASE

Empathy Map Canvas



3.REQUIREMENT ANALYSIS

3.1. DATA FLOW DIAGRAM & USER STORIES

3.1.1 User Story

The **Automating Employee Data Import System** is designed to simplify the process of importing employee data into the ServiceNow platform and automatically displaying relevant employee and managerial information during incident management.

In many organizations, employee data is stored in external systems such as spreadsheets or HR databases. Service desk agents often need employee information when assigning incidents. Manually entering or searching for this data is time-consuming and prone to errors. The proposed system automates the process of importing employee records and integrates them directly into ServiceNow.

The system enables administrators to import employee data such as name, email, department, manager name, and manager email from external files like CSV or Excel. Once the data is imported, it is automatically stored in the ServiceNow user database. When incidents are assigned to employees, the system automatically retrieves related information such as department details and manager email using dot walking.

This automation significantly reduces manual work for administrators and improves the efficiency of incident handling. Service desk agents can quickly view the department and manager details of the assigned employee directly on the incident form. This ensures faster communication and escalation when required.

Additionally, incidents typically require faster response times than other tasks. To address this, the system modifies the default priority value of the incident table using dictionary override. This ensures that incidents are assigned an appropriate priority level automatically.

By integrating automated data import, incident management, and customized priority settings, the system provides a streamlined workflow that improves productivity, data accuracy, and response time for IT support teams.

3.1.2 Data Flow Diagram (DFD)

A **Data Flow Diagram (DFD)** illustrates how data moves within the system and how different components interact to process and store information.

EXTERNAL ENTITY – EMPLOYEE DATA SOURCE

The process begins with an external data source such as a spreadsheet or HR system containing employee details. The data typically includes fields like employee name, email, department, manager name, and manager email. This data is exported into formats such as CSV, XML, or XLSX before being uploaded to the ServiceNow platform.

Process 1 – Import Set

Once the file is uploaded, ServiceNow creates an **Import Set Table**. This table acts as a temporary storage location where the imported data is stored before being processed. The import set ensures that the data can be validated and transformed before it is inserted into the main database.

Process 2 – Transform Map

After the data is loaded into the import set table, a **Transform Map** is used to map the fields from the staging table to the corresponding fields in the target table. This mapping ensures that employee attributes such as name, department, and manager information are correctly transferred to the ServiceNow user table.
The transform map also performs data validation and transformation during this process. Any mismatched or incorrect data can be identified and corrected before being inserted into the system.

Process 3 – User Table Storage

Once the transformation is completed, the employee data is inserted into the **ServiceNow User Table (sys_user)**. This table stores all user records within the platform and allows other modules such as incident management to access employee information.

Process 4 – Incident Management

When an incident is created in the system and assigned to a specific employee, the incident form retrieves relevant employee details automatically. Using dot walking, the system can display related fields such as department, employee email, and manager email directly on the incident form.
This ensures that service desk agents have immediate access to essential information required for incident resolution and escalation.

Process 5 – Dictionary Override

The incident table is derived from the task table and inherits its default properties. However, incidents typically require a different default priority level. Using the dictionary override feature, the system modifies the default priority value specifically for incidents.
This ensures that incidents receive a priority level that reflects their urgency without affecting the parent task table.

3.2 SOLUTION REQUIREMENTS

3.2.1 Functional Requirements

Functional requirements describe the core capabilities that the system must provide in order to meet user needs.
The system must allow administrators to import employee data from external sources such as Excel spreadsheets, CSV files, or XML files. The imported data should include important employee details such as name, email address, department, manager name, and manager email.
The system must create an **Import Set Table** that stores the uploaded data temporarily before it is processed. This staging table ensures that data validation and transformation can occur before the information is inserted into the main user table.
Another important requirement is the ability to create and configure **Transform Maps**. Transform maps define how fields from the staging table correspond to fields in the target table. For example, the employee name field from the import file should map to the name field in the ServiceNow user table.
The system must also support **automatic field mapping** through mapping assist. This feature helps administrators automatically match fields with similar names between the staging table and the target table, reducing configuration effort.
Another important functionality is **dot walking**, which allows related data from referenced tables to be displayed directly on forms. When an incident is assigned to an employee, the system should automatically display the department name, employee email, and manager email.
The system must also allow administrators to configure **dictionary overrides**. This feature enables modifications to inherited field properties such as default values. In this project, the default priority value of the incident table is changed to ensure that incidents receive an appropriate priority level.

3.2.2 Non-Functional Requirements

Non-functional requirements describe the performance, reliability, and quality attributes of the system.

Usability
The system should provide an intuitive and user-friendly interface that allows administrators and service desk agents to perform tasks efficiently. The import process should be simple and require minimal technical expertise. Clear form layouts and easy navigation help users perform their tasks quickly.

Security
Employee data contains sensitive information such as email addresses and departmental details. The system must ensure that only authorized users can access or modify this data. Role-based access control should be implemented to restrict access to administrative functions such as data import and transform map configuration.

Reliability
The system must operate reliably without data loss or corruption. Data import and transformation processes should execute correctly and consistently. Error handling mechanisms should identify and report issues during the import process.

Performance
The system should process large datasets efficiently without affecting the performance of the ServiceNow platform. Data imports and transformations should be completed within a reasonable time frame even when handling large numbers of employee records.

Scalability
The system should support future growth in employee data and organizational structure. As the number of employees increases, the system should continue to operate efficiently without requiring major modifications.

Availability
The system should be available to users during organizational working hours and should minimize downtime. Cloud-based deployment ensures that users can access the system from different locations.

3.3. TECHNOLOGY STACK

3.3.1 ServiceNow Platform

The primary platform used in this project is **ServiceNow**, a cloud-based platform that provides tools for IT service management and workflow automation.

ServiceNow allows organizations to automate business processes, manage incidents, and integrate data from multiple sources. The platform provides built-in features such as Import Sets, Transform Maps, and Incident Management modules that simplify system development.

By using ServiceNow, the organization can centralize employee information and automate incident management workflows.

3.3.2 Import Sets

Import Sets are used to import external data into ServiceNow. They act as an intermediate staging area where uploaded data is stored temporarily before being processed.

When an administrator uploads a file containing employee data, ServiceNow creates an import set table. Each row in the file becomes a record in the staging table. This allows administrators to review and validate the data before performing the transformation process.

3.3.3 Transform Maps

Transform Maps define how data from the staging table should be transferred to the target table.

Each field in the staging table is mapped to a corresponding field in the target table. For example, the employee email field in the import file is mapped to the email field in the ServiceNow user table.

Transform maps also support data transformation logic, allowing administrators to modify or validate data during the import process.

3.4 Dot Walking

Dot walking is a powerful feature in ServiceNow that allows users to access related fields from referenced tables without opening separate records.

In this project, dot walking is used to display employee department details, email address, and manager information directly on the incident form. This allows service desk agents to view relevant information quickly and reduces the need for manual searches.

3.5 Dictionary Override

Dictionary Override is used to modify field properties that are inherited from parent tables.

The incident table in ServiceNow extends from the task table and inherits its default field values. However, incidents often require different priority settings compared to other tasks. Using dictionary override, administrators can change the default priority value specifically for the incident table without affecting other tables.

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Organizations often face challenges when importing employee data from external systems into ServiceNow. Manual entry of large datasets can lead to errors, duplication, and inconsistent data relationships. Additionally, when incidents are assigned to employees, important information such as department or manager details may not be readily visible.

The proposed solution uses ServiceNow's **Import Sets and Transform Maps** to automate the process of importing employee records. Once the data is imported, relationships between users, departments, and managers are established. Using **Dot Walking**, related information such as department and manager email can be retrieved dynamically on forms without needing to open additional records.

This solution ensures that data is accurately transferred from external sources while maintaining proper relationships between tables. It significantly improves the efficiency of incident management and reduces manual data handling.

4.2 Proposed Solution

Parameter	Description
Problem	Manual import of employee records
Solution	Automated import using ServiceNow Import Sets
Technology	ServiceNow platform
Key Feature	Dot Walking for relationship mapping
Benefit	Faster access to employee information
Scalability	Supports growing employee datasets

4.3 Solution Architecture

The system architecture includes multiple layers that work together to automate data import and retrieval.

- **External Data Layer** – Employee data stored in spreadsheet files.
- **Import Layer** – Import Sets capture the uploaded file and store it in a staging table.
- **Transformation Layer** – Transform Maps convert staging table data into ServiceNow target tables.
- **Database Layer** – Employee records are stored in the **User (sys_user)** table.
- **Application Layer** – Incident forms retrieve related information using **Dot Walking**.
- **Configuration Layer** – Dictionary Overrides customize field defaults in extended tables.

5. PROJECT PLANNING PHASE

5.1Create records in the sheet

1.Open google sheet

2. Nuance the fields as name ensile department manager and manager's email

3. Enter some records

4. Save the sheet as employee data.

5.2 Import data

1. Open service now.

2. Click on All search for import sets

3. Select load data under system import sets

4. Select create table

5. Give Import set table name as: import employee

6. Choose the file which we created in the XL sheet.

7. Then we are attached the pages

5.3 Create Transform Map

1. Open ServiceNow.

2. Click on All and search for Import Sets.

3. Click on Transform Maps.

4. Click on New to create a Transform Map under Import Sets.

5. Fill in the following details to create a new Transform Map.

6. Give the Name as: *Employee Data Transform*.

7. Select the Source Table as: *Import Employee Data*.

8. Select the Target Table as: *User [sys_user]*.

9. Click Save to store the form.

5.4 Match the fields using mapping assist

1. Scrolled down the created transform map form
2. Click on auto matching fields
3. Then in the mapping fields some fields are matched
4. Using mapping assist map the field
5. Select the fields which you are mapping in the staging table and target table
6. Select manager in source table and add
7. Select manager field in target table
8. Click on the save button.

5.5 Transform the Data

1. Click on transform in the related links.
2. Click on transform.
3. Data's transform state is completed**.**

6. PROJECT DEVELOPMENT PHASE

6.1 PERFORMANCE TESTING

6.1.1 Project Development

1.Problem statement
2.Empathy Map Canvas
3.Brainstoring and Idea Prioritisation
4.Data flow diagram & User Stories
5.Solution Requirement
6.Tecnology Stacks

Why you'll learn

1. In the future, the project can be improved by adding more advanced features.
2. The system can be developed as a mobile application for easy access.
3. Security features can be enhanced to protect user data.
4. The user interface can be improved to make the system more user-friendly.
5. Cloud storage can be added to store and manage data efficiently.
6. The system performance can be optimized for faster processing.
7. More users can be supported with better database management.
8. Additional functions can be included based on user requirements.
9. Regular updates can be provided to improve system reliability.
10. This project can be expanded and integrated with other technologies in the future.

7. ADVANTAGES AND DISADVANTAGES

Advantages

- Reduces manual effort in data import
- Improves data accuracy and consistency
- Saves time by automating the import process
- Maintains proper relationships between tables
- Handles large volumes of data efficiently

Disadvantages

- Initial setup and configuration may take time
- Requires technical knowledge to implement
- Errors in mapping may affect data relationships

8. PROJECT DOCUMENTATION

8.1 FINAL REPORT

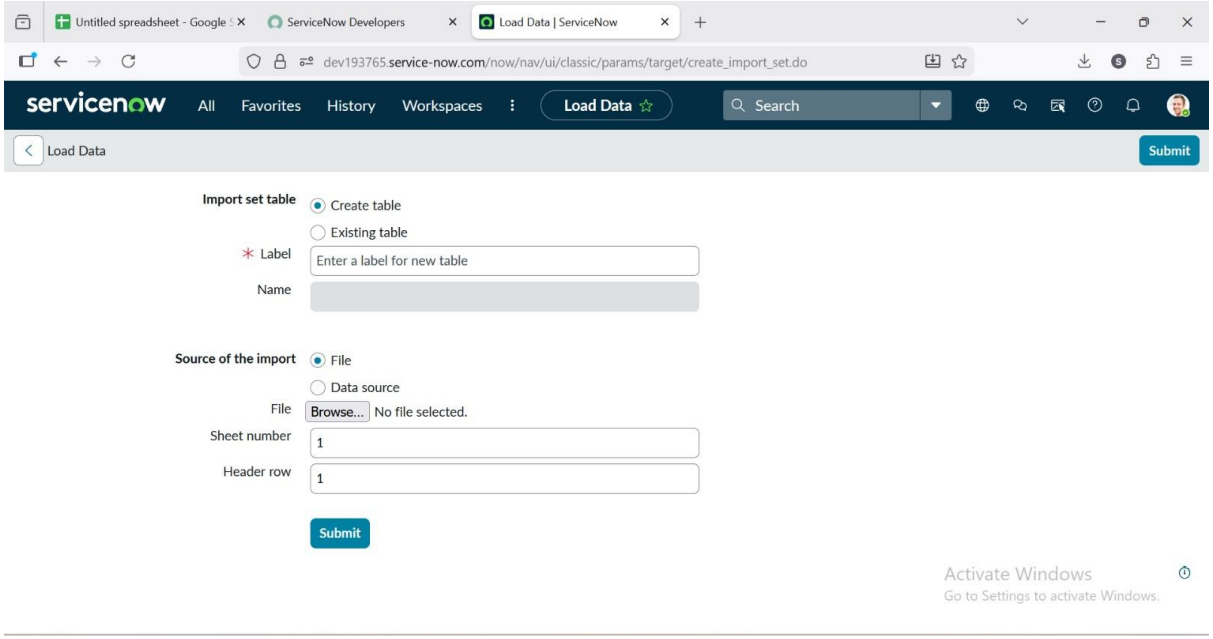
The final report provides a complete description of the project **"Automating Data Import and Relationship using Import Sets and Dot Walking using ServiceNow."** It includes details about the problem identified, the proposed solution, system design, development process, and the results achieved after implementing the system. The project was developed to automate the process of importing large volumes of data and maintaining relationships between different tables in an efficient way.

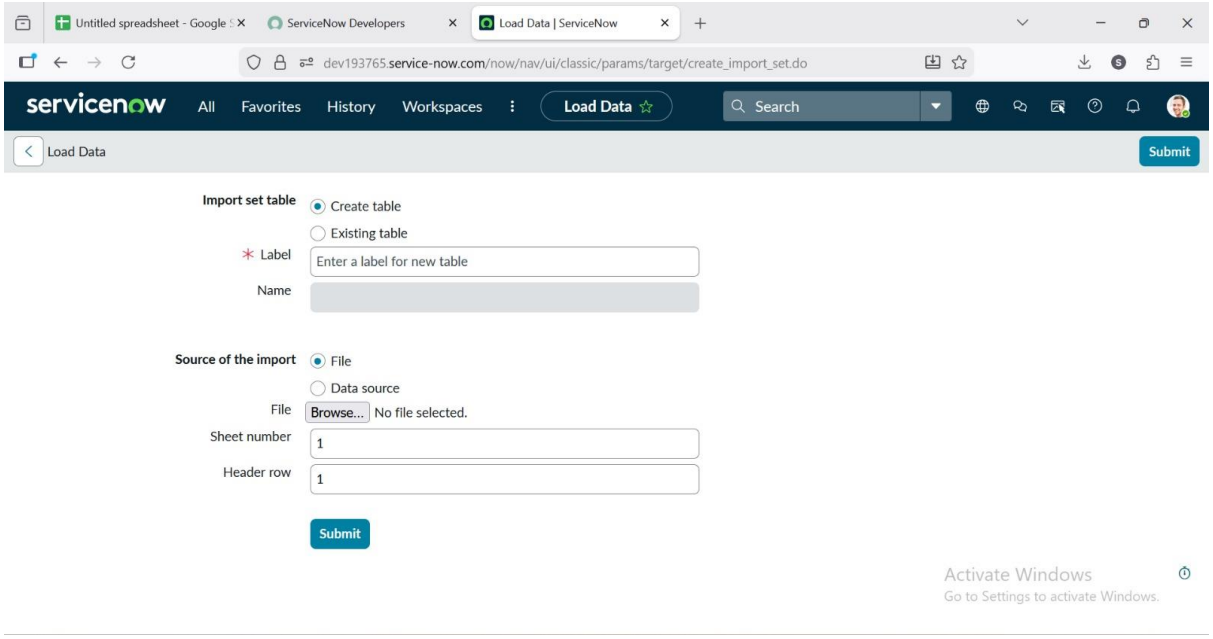
In many organizations, data is often imported manually from external sources such as spreadsheets or databases. This manual process can lead to errors, duplication of records, and difficulty in maintaining relationships between tables. To overcome these issues, an automated solution was developed using the **ServiceNow platform**. The system uses **Import Sets** to load external data into a staging table and **Transform Maps** to move the data into the target tables.

The use of **Dot Walking** helps in accessing and maintaining relationships between related records across different tables. This simplifies the process of managing linked data and improves the accuracy of the system. The project also includes different phases such as ideation, requirement analysis, project design, development, and testing to ensure that the system works correctly and meets the project objectives.

The final system successfully demonstrates how automation can simplify the process of importing and managing large volumes of data. This project highlights the importance of using modern automation techniques to improve efficiency, reduce manual effort, and maintain accurate data relationships within an organization.

8.2 SCREEN LAYOUT





Untitled spreadsheet - Google S | ServiceNow Developers | ServiceNow

dev193765.service-now.com/now/nav/ui/classic/params/target/import_status.do%3Fsysparm_pworker_

servicenow All Favorites History Admin ServiceNow Search

Progress

Name	ImportProcessor
State	Complete
Completion code	Success
Message	Processed: 4, inserts 4, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.212)

Next steps...

Import sets Go to the import sets for this data load

Loaded data Go to the newly imported data inside the staging table: u_123

Create transform map Create a transform map for the newly staged data

Import log View the import log

Activate Windows
Go to Settings to activate Windows.

Untitled spreadsheet - Google S | ServiceNow Developers | New Record | Table Transform M

dev193765.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3D-1

servicenow All Favorites History Table Transform Map - New Record Search

Table Transform Map
New record

Submit

* Name	Test	Application	Global
* Source table	@123 [u_123]	Created	
Active	☑	* Target table	User [sys_user]
Run business rules	☑	Order	100
Enforce mandatory fields	No	Run script	☐
Copy empty fields	☐		
Create new record on empty coalesce fields	☐		

Submit

Related Links

Auto Map Matching Fields

Mapping Assist

Activate Windows
Go to Settings to activate Windows.

Untitled spreadsheet - Google S | ServiceNow Developers | Test | Table Transform Map | Ser

dev193765.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3Df6c

servicenow | All | Favorites | History | Table Transform Map - Test | Search

Table Transform Map
Test

Copy | Update | Delete

Field maps created

* Name	Test	Application	Global
* Source table	@123 [u_123]	Created	2026-03-05 04:46:05
Active	☑	* Target table	User [sys_user]
Run business rules	☑	Order	100
Enforce mandatory fields	No	Run script	☐
Copy empty fields	☐		
Create new record on empty coalesce fields	☐		

Copy | Update | Delete

Related Links

Auto Map Matching Fields

Mapping Assist

Validate Coalesce Fields

Activate Windows
Go to Settings to activate Windows.

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dev193765.service-now.com/now/nav/ui/classic/params/target/slushbucket.do%3Fsysparm_refe 90%

servicenow | All | Favorites | History | Workspaces | ServiceNow | Search

Language	State
Last login	On hold reason
Last login time	Impact
Last name	Urgency
Location [+]	Priority
Locked out	Assignment group
Manager [+]	Assigned to
Manager HP1	Assigned to.Department
Middle name	Assigned to.Email
Mobile phone	Assigned to.Manager
Name	|- end_split -|
Notification	Short description
Password	Description
Password needs reset	Contextual Search Results

Cancel | Save

Form view and section

View name: Default view

Section: Incident, Notes, Related Records, Resolution Information

Create new field

Name

Type: String

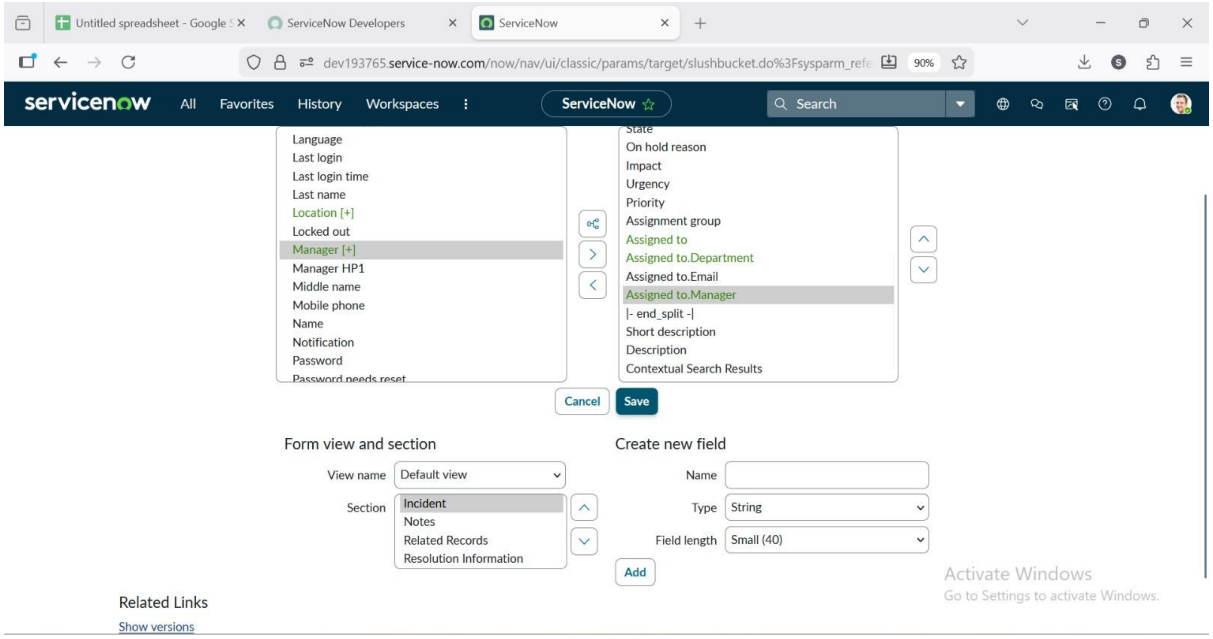
Field length: Small (40)

Add

Activate Windows
Go to Settings to activate Windows.

Related Links

Show versions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ServiceNow Developers
ServiceNow
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Favorites
History
Workspaces
ServiceNow
Search
Language
Last login
Last login time
Last name
Location [+]
Locked out
Manager [+]
Manager HP1
Middle name
Mobile phone
Name
Notification
Password
Password needs reset
State
On hold reason
Impact
Urgency
Priority
Assignment group
Assigned to
Assigned to.Department
Assigned to.Email
Assigned to.Manager
|- end_split -|
Short description
Description
Contextual Search Results
Cancel
Save
Form view and section
View name
Default view
Section
Incident
Notes
Related Records
Resolution Information
Create new field
Name
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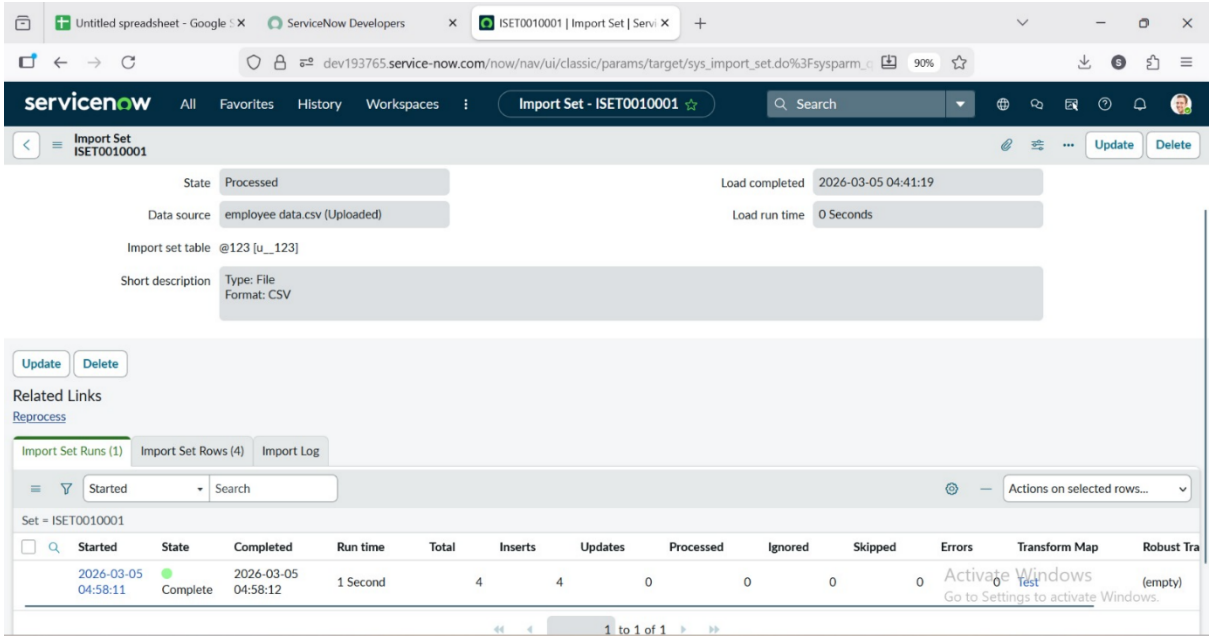
String
Field length
Small (40)
Add
Activate Windows
Go to Settings to activate Windows.
Related Links
Show versions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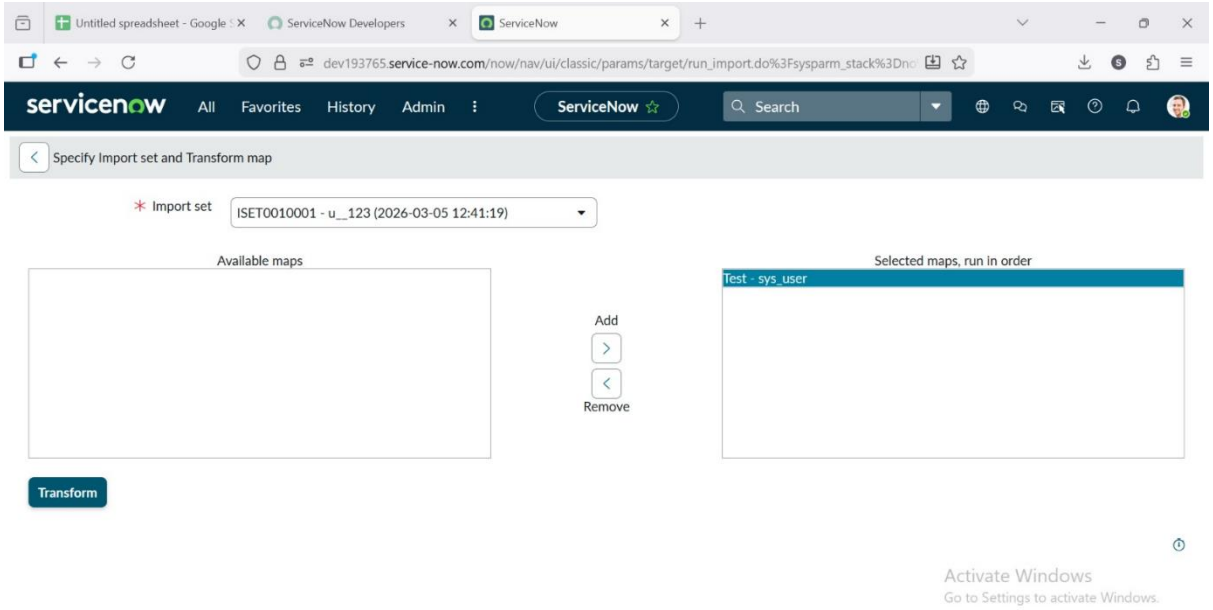
90%
servicenow
All
Favorites
History
Workspaces
Import Set - ISET0010001
Search
Import Set
ISET0010001
Update
Delete
State
Processed
Load completed
2026-03-05 04:41:19
Data source
employee data.csv (Uploaded)
Load run time
0 Seconds
Import set table
@123 [u_123]
Short description
Type: File
Format: CSV
Update
Delete
Related Links
Reprocess
Import Set Runs (1)
Import Set Rows (4)
Import Log
Started
Search
Actions on selected rows...
Set = ISET0010001
Started
State
Completed
Run time
Total
Inserts
Updates
Processed
Ignored
Skipped
Errors
Transform Map
Robust Tra
2026-03-05 04:58:11
Complete
2026-03-05 04:58:12
1 Second
4
4
0
0
0
0
0
Test
(empty)
Activate Windows
Go to Settings to activate Windows.
1 to 1 of 1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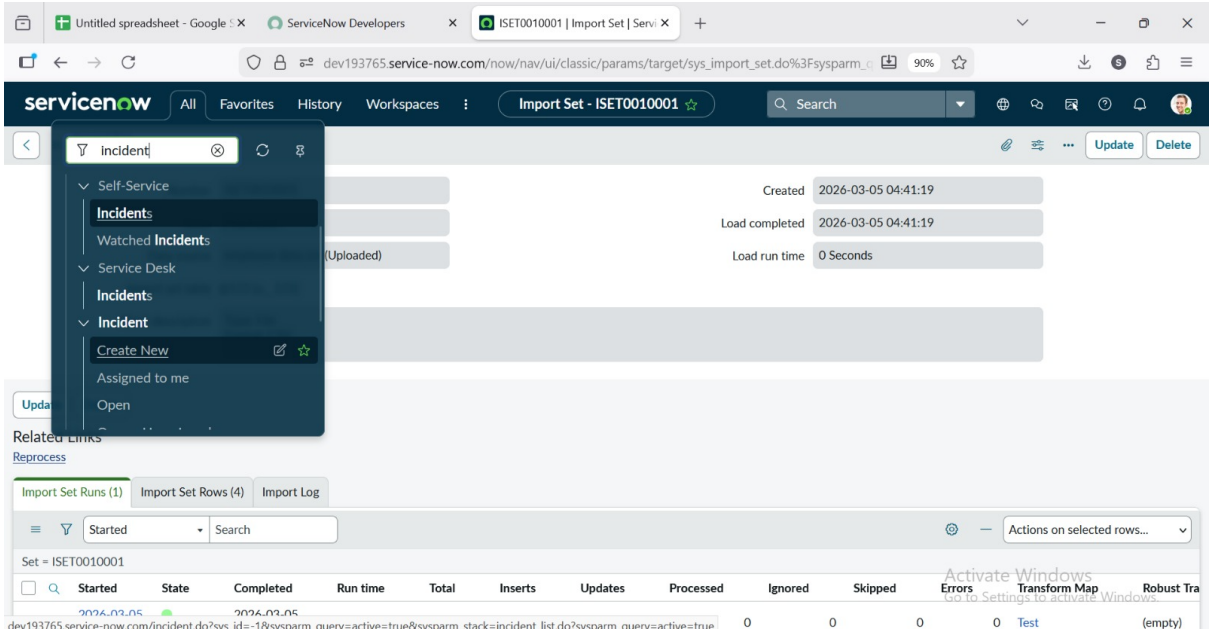
Favorites
History
Admin
ServiceNow
Search
Specify Import set and Transform map
Import set
ISET0010001 - u__123 (2026-03-05 12:41:19)
Available maps
Add
Remove
Selected maps, run in order
Test - sys_user
Transform
Activate Windows
Go to Settings to activate Windows.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90%
servicenow
All
Favorites
History
Workspaces
Import Set - ISET0010001
Search
Update
Delete
incident
Self-Service
Incidents
Watched Incidents
Service Desk
Incidents
Incident
Create New
Assigned to me
Open
Created
2026-03-05 04:41:19
Load completed
2026-03-05 04:41:19
Load run time
0 Seconds
(Uploaded)
Related Links
Reprocess
Import Set Runs (1)
Import Set Rows (4)
Import Log
Started
Search
Actions on selected rows...
Set = ISET0010001
Started
State
Completed
Run time
Total
Inserts
Updates
Processed
Ignored
Skipped
Errors
Transform Map
Robust Tra
0
0
0
0
Test
(empty)
Activate Windows
Go to Settings to activate Windows.

9. CONCLUSION

The project **“Automating Data Import and Relationship using Import Sets and Dot Walking”** was successfully implemented using the ServiceNow platform. The main objective of this project was to automate the process of importing data from external sources and maintain proper relationships between different tables. By using Import Sets and Transform Maps, data can be imported efficiently into the system while reducing manual effort and minimizing errors. The use of Dot Walking helps in easily accessing related data across multiple tables and maintaining accurate relationships within the database.

This project demonstrates how automation can improve data management and increase efficiency in organizations that handle large volumes of data. The system reduces the time required for manual data processing and ensures better data accuracy and consistency. Overall, the implementation of this automated solution highlights the importance of modern tools and techniques in simplifying complex data operations and improving system performance.

PROJECT PHASES OVERVIEW

Automating Data Import & Relationship using Import Sets and Dot Walking

IDEATION PHASE

- Understand the problem of manual data import.
- Brainstorm ideas for an automated solution.

REQUIREMENT ANALYSIS

- Define data sources, fields, and relationships.
- Identify necessary transformations and mappings.

PROJECT DESIGN PHASE

- Design the Import Sets and Transform Maps.
- Plan the Dot Walking paths for linking relations.

PROJECT PLANNING PHASE

- Organize tasks and allocate resources.
- Set timelines for successful implementation.

PROJECT PLANNING PHASE

- Organize tasks and allocate resources.
- Set timelines for successful implementation.

PROJECT DEVELOPMENT PHASE

- Develop and configure the automated system.
- Test data import and relationship mappings.

FINAL REPORT

- Complete a detailed report covering the project.
- Summarize results and present key findings.